



CS302- Digital Logic Design
(SOLVED MCQs)
FROM MIDTERM PAPERS
LECTURE (1-22)



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CS619 & CS519

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1. The 4-variable K-Map has _____ rows and _____ columns of cells.
 - ☒ 4,4
 - ☐ 2,2
 - ☐ 3,3
 - ☐ 6,6
2. The switching speed of CMOS is now _____.
 - ☒ competitive with TTL
 - ☐ three times that of TTL
 - ☐ slower than TTL
 - ☐ twice that of TTL
3. Gray code of 8 is:
 - ☒ 1100
 - ☐ 1000
 - ☐ 0100
 - ☐ 0000
4. Which circuit is used in 7-segments display?
 - ☒ BCD Code
 - ☐ Excess Code
 - ☐ Binary code
 - ☐ Gray code
5. The 4-bit 2's complement representation of "-7" is ____
 - ☐ 0111
 - ☐ 1111
 - ☒ 1001
 - ☐ 0110
6. Don't care conditions are marked as _____ in the output column of the function table
 - ☐ A
 - ☐ 1
 - ☒ X
 - ☐ 0
7. The digital circuits operate with which combination of values given below?
 - ☐ 0 volts, 0 volts
 - ☒ +5 volts, 0 volts
 - ☐ -5 volts, 0 volts

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-  -5 volts, 0 volts
8. In 15-digit decimal floating-point format, the biased exponent value _____ can be used to represent the number zero whatever the value of the mantissa.
-  00
-  000
-  0
-  0000
9. The ANSI/IEEE Standard 754 defines a _____ Single-Precision Floating Point format for binary numbers.
-  16-bit
-  32-bit
-  64-bit
-  8-bit
10. A function represented by an expression in _____ form can be readily programmed.
-  Boolean
-  Standard POS
-  Standard SOP
-  Non-standard SOP
11. _____ is an essential part of sequential circuit.
-  Clock
-  Memory element
-  JK flip flop
-  Multiplexer
12. The OR Gate performs a Boolean _____ function
-  Multiplication
-  Addition
-  Subtraction
-  Division
13. which of the following is the hexadecimal equivalent of 28?
-  1A
-  1C
-  1B
-  1D
14. To implement the decoder circuit, having _____ inputs and _____ outputs, function tables have to be drawn which represent the output status of each output line for all combinations of inputs.
-  3 and 8
-  4 and 7
-  1 and 9
-  5 and 6

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15. The _____ Gate is used in applications where the output signal is a 1 when any one input is a 1.

☒ OR

☐ XOR

☐ NOT

☐ AND

16. “1101” in signed representation is equivalent to _____

☐ 13

☒ -5

☐ 10

☐ -10

17. In a 4-variable K-map, a 2-variable product term is produced by

☐ A 4-cell group of 0s

☐ A 8-cell group of 1s

☒ A 4-cell group of 1s

☐ 2-cell group of 1s

18. Suppose we want to transmit the data “10001101”, and an “Even-Parity” bit scheme is used to detect errors, the parity bit added to this data will be _____

☒ “0”

☐ “1”

☐ Both “0” and “1”

☐ Parity bit is not needed in this case

19. A variable can have a _____ value.

☐ 0

☒ 0 or 1

☐ 0 and 1

☐ 1

20. XOR is an abbreviation of _____.

☐ Inclusive-OR

☐ Extended-OR

☐ Extendable-NOR

☒ Exclusive-OR

21.

Inputs		Output
A	B	F
0	0	0
0	1	1
1	0	1
1	1	1

This function table represents _____ Gate.

☒ NAND

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☐ AND

☒ OR

☐ XOR

22. Which of the following is the example of comparator?

☐ AND

☐ XOR

☐ OR

☒ XNOR

23. The decimal value “20” is equivalent to binary value _____.

☐ 10011

☐ 11001

☐ 00101

☒ 10100

24. Boolean algebra is the _____ of Digital Systems.

☐ Science

☐ Equation

☐ Expression

☒ Mathematics

25. In 15-digit decimal floating-point format, the biased exponent value _____ can be used to represent the number infinity whatever the value of mantissa.

☒ 99

☐ 98

☐ 256

☐ 255

26. The digital circuits operate with which combination of values given below?

☐ A=0, B=0, C=1, D=1

☒ A=1, B=0, C=0, D=1

☐ A=0, B=1, C=1, D=0

☐ A=1, B=1, C=0, D=0

27. A.....is converted into a standard POS by using the rule $A\bar{A}=0$.

☐ No-standard SOP

☒ Non-standard POS

☐ Simplified POS

☐ Simplified SOP

28. $(A+B).(A+C)=$ _____

☒ A+BC

☐ B+C

☐ AC+B

☐ AB+C

29. The PLDs are implemented having a general-purpose structure based on.....arrays

☐ AND+NANAD

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☒ AND-OR

☐ XOR-NOR

☐ OR-XOR

30. Each octal Number digit can represent a.....binary Number.

☐ 2-bit

☒ 3-bit

☐ 4-bit

☐ 7-bit

31. The output of any CMOS.....series logic gate can be at logic high “1” or logic low “0”.

☐ 3.3 and 5 volt

☐ 5+ volt

☐ 4.5 volt

☒ 5 volt

32. A standard SOP from has.....terms that have all the variables in the domain of the expression.

☐ Sum

☐ Max

☐ Composite

☒ Product

33. In caveman number system the value “2” is represented by Symbol_____.

☒ “>”

☐ “!”

☐ “^”

☐ “&”

34. The expression $F=A.B.C$ describes the Operating of three bits.....Gate.

☐ NAND

☐ NOT

☐ OR

☒ AND

35. In CMOS 5 volts series, input voltage of logic high signal (V_H) with a range from.....To..... volts.

☐ 0, 3.5

☒ 4.5,5

☐ 0,5

☐ 3.5,5

36. Sum term (Max term) is implemented using.....gates.

☒ NOT

☐ AND

☐ NOR

☐ OR

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37. The output $A < B$ is set to 1 when the input combinations is _____

✚ A=10, B=01

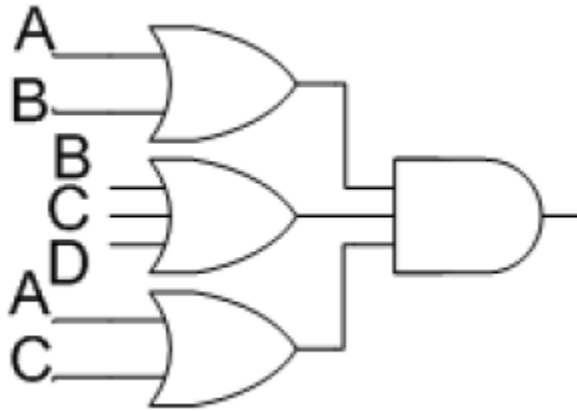
✚ A=11, B=01

✚ A=01, B=01

✚ A=01, B=10

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38. The diagram given below represents.....



✚ Demorgan's law

✚ Associative law

✚ Product of sum form (According to rule of theorem)

✚ Sum of product form

39. A BCD to 7-Segment decoder has

		AB				
		00	01	11	10	
CD	00	1	1	1	0	A
	01	1	0	0	1	B
	11	0	1	1	1	C
	10	0	1	1	0	D

✚ 3 inputs and 7 outputs

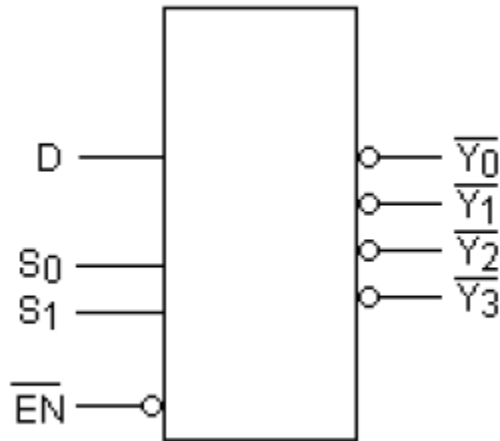
✚ 4 inputs and 7 outputs (Page 103)

✚ 7 inputs and 3 outputs

✚ inputs and 4 outputs

40. The device shown here is most likely a _____

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- ✦ Comparator
- ✦ Multiplexer
- ✦ **Demultiplexer**
- ✦ Parity generator

41. A Karnaugh map is similar to a truth table because it presents all the possible values of input variables and the resulting output of each value.

- ✦ **True**
- ✦ False

42. Generally, the Power dissipation of devices remains constant throughout their operation.

- ✦ **TTL (Page 65)**

- ✦ CMOS 3.5 series
- ✦ CMOS 5 Series

- ✦ **Power dissipation of all circuits increases with time.**

43. The OLMC of the GAL16V8 is _____ to the OLMC of the GAL22V10

- ✦ Similar
- ✦ Different
- ✦ **Similar with some enhancements (Page 207)**
- ✦ Depends on the type of PALs input size

44. A latch has _____ stable states

- ✦ One
- ✦ **Two (Page 218)**
- ✦ Three
- ✦ Four

45. A Demultiplexer is not available commercially.

- ✦ **True (Page 178)**
- ✦ False

46. A logic circuit with an output $X = \bar{A}BC + A\bar{B}$ consists of _____

- ✦ two AND gates, two OR gates, two inverters
- ✦ three AND gates, two OR gates, one inverter
- ✦ **two AND gates, one OR gate, two inverters**
- ✦ two AND gates, one OR gate

47. Following is standard POS expression

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$$(A + \bar{B} + C + \bar{D})(A + \bar{B} + C + D)(A + B + \bar{C} + \bar{D})(A + B + C + \bar{D})(A + \bar{B} + \bar{C} + D)$$

☒ True (According to logic)

☐ False

48. $\bar{A}B + \bar{A}B\bar{C} + AC$ is an example of _____

☐ Product of sum form

☒ Sum of product form (Page 77)

☐ Demorgans law

☐ Associative law

49. Two 2-input, 4-bit multiplexers 74X157 can be connected to implement a _____ multiplexer.

☐ 4-input, 8-bit

☐ 4-input, 16-bit

☐ 2-input, 8-bit

☒ 2-input, 4-bit (Page 169)

50. The PROM consists of a fixed non-programmable _____ Gate array configured as a decoder.

☒ AND (Page 182)

☐ OR

☐ NOT

☐ XOR

51. The range of Excess-8 code is from.....to.....

☒ +7 to -8

page 34

☐ +8 to -7

☐ 9 to -8

☐ -9 to +8

52. The decimal "17" in BCD will be represented as.....

☐ 11101

☐ 11011

☒ 10111

☐ 11110

53. A started interface for programming the In-System PLD consists of.....

☐ 2-wire

☒ 4-wire

☐ 8-wire

☐ 16-wire

54. Dual, 4-input multiplexer can be connected to from a 16-input multiplexer.

☐ 2

☐ 3

☒ 4

☐ 5

55. A product term is 0 when.....

☐ Any of the literals is 1

☐ At least two literals are 1

☐ All the literals are 1

☒ Any of the literals is 0

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56. What will be the output of a 3-input NAND gate for the input values A=1, B=0, and C=1.

- ☒ Zero
- ☐ One
- ☐ Undefined
- ☐ No output, as input is invalid

57.is not a valid hexadecimal number.

- ☐ 1234
- ☐ ABCD
- ☐ 1001
- ☒ DEHF

58. ABCD to 7-segment decoder has.....inputs andoutputs.

- ☐ 3, 7
- ☒ 4, 7
- ☐ 7, 3
- ☐ 7, 4

59. Which of the following simplification method is intended to be used for Boolean expressions having more than four variables?

- ☐ Boolean Algebra and rules
- ☐ Karnaugh Map
- ☒ Quine-McCluskey
- ☐ DE Morgan's Theorem

60. What are the values of sum and carry out when two 4-bit binary numbers (1011 and 1111) are applied to a 4-bit parallel adder and carry input is 1.

- ☐ $\sum_4 \sum_3 \sum_2 \sum_1 = 0111, C_{aul} = 0$
- ☐ $\sum_4 \sum_3 \sum_2 \sum_1 = 1111, C_{aul} = 1$
- ☐ $\sum_4 \sum_3 \sum_2 \sum_1 = 1011, C_{aul} = 1$
- ☐ $\sum_4 \sum_3 \sum_2 \sum_1 = 1100, C_{aul} = 1$

61. A particular half-adder has.....inputs and.....output(s).

- ☐ 3, 1
- ☐ 3, 2
- ☐ 2, 1
- ☒ 2, 2

62. TTL based device work with a DC supply of.....Volts.

- ☐ +10
- ☒ +5
- ☐ +3
- ☐ 3.3

63. In CMOS 5 Volt series, input voltage for logic high signal (V_{ih}) is in the range of.....volts.

- ☒ 3.5 to 5
- ☐ 4.5 to 5
- ☐ 0 to 5
- ☐ 0 to 3.5

64. A multiplexer circuit has.....input(s) and.....output(s).

- ☐ Single, single
- ☐ Single, multiple
- ☐ Multiple, single

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✚ Multiple, multiple

65. The binary values for the standard SOP expression, $\bar{A}BCD + \bar{A}BC\bar{D} + \bar{A}\bar{B}\bar{C}D$ are.....

✚ 1110 + 0110 + 0001

✚ 1011 + 1111 + 1011

✚ 0001 + 1001 + 1110

✚ 1010 + 1110 + 0101

66. Determine the binary values of the variables for the following standard POS expression.
 $(A + \bar{B} + C)(\bar{A} + B + \bar{C})$

✚ (1+0+1)(0+1+0)

✚ (0+1+0)(1+0+1)

✚ (1+0+0)(0+0+1)

✚ (0+1+0)(1+1+0)

67. The first Least Significant digit in decimal number system has,

✚ Position 0 and weight equal to 1

✚ Position 1 and weight equal to 0

✚ Position 1 and weight equal to 10

✚ Position 0 and weight equal to 10

68. Consider the binary number “100101”, Its 2’s complement will be.....

✚ 100110

✚ 011011

✚ 100111

✚ 011010

69. $A + \bar{A}B = \dots\dots\dots$

✚ A

✚ B

✚ A B

✚ A + B

70. According to Dem organ’s theorem:

$\bar{A} + \bar{B} + \bar{C} = \dots\dots\dots$

✚ A B C

✚ $A + \bar{B}\bar{C}$

✚ $\bar{A} \cdot \bar{B} \cdot \bar{C}$

✚ $\bar{A}\bar{B} + C$

71. The three fundamental gates are.....

✚ AND, NAND, XOR

✚ OR, AND, NAND

✚ NOT, NOR, XOR

✚ NOT, OR, AND

72. The output $A < B$ is set to 1 when the input condition is.....

✚ A=10, B=01

✚ A=11, B=01

✚ A=01, B=01

✚ A=01, B=10

73. An alternate method of implementing Comparators which allows the Comparators to be easily cascaded without the need for extra logic gates is.....

✚ Using a single comparator

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✚ Using Iterative Circuit based comparators

✚ Connecting comparators in vertical hierarchy

✚ Extra logic gates are always required

74. Two 2-input, 4-bit multiplexers 74×157 can be connected to implement a.....multiplexer.

✚ 2-input, 8-bit

✚ 2-input, 4-bit

✚ 4-input, 8-bit

✚ 4-input, 16-bit

75. $(A+B)(A+\bar{B}+C)(\bar{A}+C)$ is an example of.....

✚ Product of sum form

✚ Sum of product form

✚ Demorgan's Law

✚ Associative Law

76. The expression $F=A+B+C$ describe the operation of 3-bit.....gate

✚ OR

✚ AND

✚ NOT

✚ NAND

77. Which of the following is a symbol of OR operation in ABEL?

✚ #

✚ \$

✚ !

✚ &

78. A latch retains its state unless.....

✚ Power is turned off

✚ Input is changed

✚ Output is changed

✚ Clock plus is changed

79. At $S=0$ and $R=1$, an active-HIGH SR latch is in.....condition.

✚ SET

✚ RESET

✚ Invalid

✚ No change

80. What are the sum (Σ) and the carry (C_{out}) for full adder circuit when $A=1$, $B=1$ and $C_{in}=0$?

✚ $\Sigma=0$, $C_{aul}=1$

✚ $\Sigma=1$, $C_{aul}=0$

✚ $\Sigma=1$, $C_{aul}=1$

✚ $\Sigma=0$, $C_{aul}=0$

81. Consider the binary number "100101", its 2's complement will be.....

✚ 100110

✚ 011011

✚ 100111

✚ 011010

82. The values that exceed the specified range can't be correctly represented and are considered as.....

✚ Overflow

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- ☐ Carry
 - ☐ Parity
 - ☐ Sign value
83. The output of an AND gate is one when.....
- ☒ All of the inputs are one
 - ☐ All of the inputs are zero
 - ☐ Any of the input is one
 - ☐ Any of the inputs is zero
84. A standard SOP form has.....terms that have all the variables in domain of the expression.
- ☐ Sum
 - ☒ Product
 - ☐ Max
 - ☐ Composite
85. A comparator circuit has multiple inputs and.....outputs.
- ☐ 2
 - ☒ 3
 - ☐ 4
 - ☐ 7
86. The active high and active low inputs of 3-to-8 Decoder are as follows:
- ☒ One active-high and remaining two are active-low.
 - ☐ Two active-high and remaining one is active-low.
 - ☐ All active-high
 - ☐ All active-low
87. OLMC consists of a.....
- ☐ Encoder
 - ☐ Decoder
 - ☐ Logic Gate
 - ☒ Flip-Flop
88. The Quad Multiplexer has.....inputs.
- ☐ 4
 - ☐ 8
 - ☐ 12
 - ☒ 16
89. The carry propagation can be expressed as.....
- ☐ $C_p = AB$
 - ☐ $C_p = A+B$
 - ☐ $C_p = A \oplus B$
 - ☐ $C_p = A + \bar{B}$
90. In the binary number “1100.111”, the first bit at the right of decimal point has the weight.
- ☐ $\frac{1}{2}$ (1 divided by 2)
 - ☐ $\frac{1}{4}$ (1 divided by 4)
 - ☐ 0
 - ☐ 1
91. Programmable Array Logic (PAL) has.....input(s) and.....output(s).
- ☐ Single, multiple
 - ☐ Multiple, single

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✚ Single, single

✚ Multiple, multiple

92. An equivalent ABEL notation of Boolean expression $A+B$ is.....

✚ $A \& B$

✚ $A | B$

✚ $A \# B$

✚ $A \$ B$

93. Which circuit has the following Sum and CarryOut expression.

Sum = $A \oplus B \oplus C$ Type equation here.

CarryOut = $C(A \oplus B) + AB$

✚ Half Adder

✚ Full Adder

✚ 3-bit parallel adder

✚ MSI adder circuit

94. From the truth table below, determine the standard SOP expression.

Inputs	Output
A B C	X
0 0 0	0
0 0 1	1
0 1 0	0

✚ $\bar{A} \bar{B} C + \bar{A} B \bar{C} + A B \bar{C}$

✚ $A B \bar{C} + A \bar{B} \bar{C} + \bar{A} \bar{B} C$

✚ $(A + B + \bar{C})(A + \bar{B} + \bar{C})(\bar{A} + \bar{B} + C)$

✚ $(\bar{A} + \bar{B} + C)(\bar{A} + B + C)(A + B + \bar{C})$

95. $(A + B)(A + C) = \dots\dots\dots$

✚ $B+C$

✚ $A+B+C$

✚ $AB+C$

✚ $AC+B$

96. Excess-8 code assigns.....to "-8".

✚ 1110

✚ 1100

✚ 1000

✚ 0000

97. The output of the expression $F+A.B.C$ will be logic....., $A=1, B=0, C=1$.

✚ Undefined

✚ One

✚ Zero

✚ No output as input is invalid

98. The cell marked 6 in 4-variable K-Map represent minterm 6 or the maxterm 6 having the following binary value of variables A,B,C and D.

✚ $A=1, B=1, C=0, D=0$

✚ $A=0, B=1, C=1, D=0$

✚ $A=0, B=0, C=1, D=1$

✚ $A=1, B=0, C=0, D=1$

99. The.....Encoder is used as a keypad encoder.

✚ 2-to-8 encoder

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- ☐ 4-to-16 encoder
 - ☐ BCD-to-Decimal
 - ☒ Decimal-to-BCD priority
100. Using multiplexer as parallel to serial converter requires.....connected to the multiplexer.
- ☒ A parallel to serial converter circuit
 - ☐ A counter circuit
 - ☐ A BCD to Decimal decoder
 - ☐ A2-to-8-bit decoder
101. In the binary number "10011" the weight of the most significant digit is.....
- ☒ 2^4 (2 raise to power4)
 - ☐ 2^3 (2 raise to power4)
 - ☐ 2^0 (2 raise to power 0)
 - ☐ 2^1 (2 raise to power 1)
102. The output of the expression $F=A+B+C$ will be logic....., When $A=0, B=1, C=1$.
- ☐ Undefined
 - ☐ One
 - ☐ Zero
 - ☒ 10 (binary)
103. A quantity having continuous value is a/an.....quantity.
- ☐ Digital
 - ☐ Binary
 - ☒ Analog
 - ☐ Natural
104. Which of the following expressions is an example of Commutative Law for Multiplication?
- ☐ $AB+C = A+BC$
 - ☐ $A(B+C) = B(A+C)$
 - ☒ $AB = BA$
 - ☐ $A+B = B+A$
105. When four 1's are taken as a group on a Karnaugh map, the number of variables eliminated from the output expression is/are.....
- ☐ 1
 - ☐ 2
 - ☐ 3
 - ☐ 4
106. Identify the Boolean expression that is in standard POS from.
- ☒ $(A + \bar{B})(\bar{A} + B + \bar{C})$
 - ☐ $\bar{A} \bar{B} C + A \bar{B} C$
 - ☐ $(A + \bar{B} + \bar{C})(A + B + C)$
 - ☐ $(A + B)(A + C)$
107. Simplifying POS (Product of sum) expression requires the grouping of.....
- ☒ 0s
 - ☐ 1s
 - ☐ Don't care
 - ☐ Both 0s and 1s
108. A SOP expression is equal to 1.....

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- ☐ All the variables in domain of expression are present
 - ☐ At least one the variables in domain of expression is present
 - ☐ When one or more product terms in the expression are equal to 0.
 - ☒ When one or more product terms in the expression are equal to 1. Page 86
109. Two 2-bit comparator circuits can be connected to form single 4-bit comparator.
- ☒ True page 154
 - ☐ False
110. High level Noise Margins (VNH) of CMOS 5 volt series circuits is.....
- ☐ 0.3 V
 - ☐ 0.5 V
 - ☒ 0.9 V page 65
 - ☐ 3.3 V
111. If we multiply “723” and “34” by representing them in floating point notation i.e. by first, converting them in floating point representation and then multiplying them, the value of mantissa of result will be.....
- ☒ 24.582 But not sure
 - ☐ 2.4582
 - ☐ 24582
 - ☐ 0.24582
112. If an active-HIGH S-R latch has a 0 on the S input and a 1 on the R input and then the R input goes to 0, the latch will be.....
- ☒ SET page 220
 - ☐ RESET
 - ☐ Clear
 - ☐ Invalid
113. 3.3 v CMOS series is characterized by.....andas compared to the 5 v CMOS series.
- ☐ Low switching speeds, high power dissipation
 - ☐ Fast switching speeds, high power dissipation
 - ☒ Fast switching speeds, very low power dissipation page 61
 - ☐ Low switching speeds, very low power dissipation
114. The binary value “1010110” is equivalent to decimal.....
- ☒ 86 (according to binary formula)
 - ☐ 84
 - ☐ 88
 - ☐ 89
115. How many data select lines are required for selecting eight inputs?
- ☐ 1
 - ☐ 2
 - ☒ 3
 - ☐ 4
116. The Quad Multiplexer has _____ outputs.
- ☒ 4 page 217
 - ☐ 8
 - ☐ 12
 - ☐ 16
117. Demultiplexer has

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- ✚ Single input and single outputs.
- ✚ Multiple inputs and multiple outputs.
- ✚ Single input and multiple outputs. Page 178)
- ✚ Multiple inputs and single output.
- 118. "Sum-of-Weights" method is used.....
 - ✚ to convert from one number system to other Page 14
 - ✚ to encode data
 - ✚ to decode data
 - ✚ to convert from serial to parallel data
- 119. The maximum number that can be represented using unsigned octal system is.....
 - ✚ 1
 - ✚ 7 page 31
 - ✚ 9
 - ✚ 16
- 120. If we add "723" and "134" by representing them in floating point notation i.e. by first, converting them in floating point representation and then adding them, the value of exponent of result will be.....
 - ✚ 0
 - ✚ 1
 - ✚ 2 page 26
 - ✚ 3
- 121. A non-standard POS is converted into a standard POS by using the rule.....
 - ✚ $A + \bar{A} = 1$
 - ✚ $A\bar{A} = 0$ Page 85)
 - ✚ $1 + A = 1$
 - ✚ $A + B = B + A$
- 122. A particular Full Adder has
 - ✚ 3 inputs and 2 output Page 135
 - ✚ 3 inputs and 3 output
 - ✚ 2 inputs and 3 output
 - ✚ 2 inputs and 2 output
- 123. The function to be performed by the processor is selected by set of inputs known as.....
 - ✚ Function Select Inputs Page 147
 - ✚ Micro Operation selectors
 - ✚ OPCODE Selectors
 - ✚ None of given option
- 124. For a 3-to-8 decoder how many 2-to-4 decoders will be required?
 - ✚ 2 Page 160
 - ✚ 1
 - ✚ 3
 - ✚ 4
- 125. GAL is an acronym for.....
 - ✚ Giant Array Logic
 - ✚ General Array Logic Page 183
 - ✚ Generic Array Logic

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- Generic Analysis Logic
126. $A.(B.C) = (A.B).C$ is an expression of.....
- DE Morgan's Law
 - Distributive Law
 - Commutative Law
 - Associative Law Page 72
127. 2's complement of any binary number can be calculated by
- adding 1's complement twice
 - adding 1 to 1's complement Page 144
 - subtracting 1 from 1's complement.
 - calculating 1's complement and inverting Most significant bit
128. The binary value "1010110" is equivalent to decimal.....
- 86 (According to formula)
 - 87
 - 88
 - 89
129. Tri-State Buffer is basically a/an.....gate.
- AND
 - OR
 - NOT
 - XOR Page 186
130. MIDTERM EXAMINATION 2010 1. The binary value "11011" is equivalent to.....
- 1B (According to rule)
 - 1C
 - 1D
 - 1E
131. An important application of AND Gate is its use in counter circuit
- True Page 281
 - False
132. The OR Gate performs a Boolean..... function
- Addition Page 42
 - Subtraction
 - Multiplication
 - Division
133. A standard POS form has _____ terms that have all the variables in the domain of the expression.
- Sum Page 85
 - Product
 - Min
 - Composite
134. A SOP expression having a domain of 3 variables will have a truth table having.....combinations of inputs and corresponding output values.
- 2
 - 4
 - 8 (According to rule)
 - 16

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135. 7. A BCD to 7-Segment decoder has
☐ 3 inputs and 7 outputs
☒ 4 inputs and 7 outputs Page 103
☐ 7 inputs and 3 outputs
☐ inputs and 4 outputs
136. In the Karnaugh map shown above, which of the loops shown represents a legal grouping?
☐ A
☐ B
☒ C
☐ D
137. The binary value of 1010 is converted to the product term
☐ True
☒ False
138. The binary numbers $A = 1100$ and $B = 1001$ are applied to the inputs of a comparator. What are the outputs levels?
☐ $A > B = 1, A < B = 0, A = B = 1$
☐ $A > B = 0, A < B = 1, A = B = 0$
☒ $A > B = 1, A < B = 0, A = B = 0$ Page 109
☐ $A > B = 0, A < B = 1, A = B = 1$
139. $C_{out1} + S_3(S_2 + S_1)$ is boolean expression for
☐ Half Adder
☐ Full Adder
☒ The Invalid BCD Detector Circuit page 142
☐ Parity Checker
140. 3-to-8 decoder can be used to implement Standard SOP and POS Boolean expressions
☒ True Page 160
☐ False
141. The GAL22V10 has.....inputs
☒ 22 Page 195
☐ 10
☐ 44
☐ 20
142. Consider a circuit consisting of two consecutive NOT gates, the entire circuit belongs to a CMOS 5 Volt series, if certain voltage is applied on the input, the output voltage of Logic high signal (VoH) will be in the range of.....volts.
☐ 4 to 4.5
☒ 4.5 to 5
☐ 0 to 4.5
☐ 0 to 3.5
143. The 4-bit 2's complement representation of "+5" is.....
☐ 1010
☐ 1110
☒ 1011
☐ 0101
144. Which of the number is not a representative of hexadecimal system

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- ☐ 1234
 - ☐ ABCD
 - ☐ 1001
 - ☒ DEHF (Hexa does not have H as remainder)
145. If we add “723” and “134” by representing them in floating point notation i.e. by first, converting them in floating point representation and then adding them, the value of exponent of result will be.....
- ☐ 0
 - ☐ 1
 - ☒ 2 Page 26
 - ☐ 3
146. The 3-variable Karnaugh Map (K-Map) has.....cells for min or max terms
- ☐ 4
 - ☒ 8 Page 89
 - ☐ 12
 - ☐ 16
147. If “1110” is applied at the input of BCD-to-Decimal decoder which output pin will be activated:
- ☐ 2nd
 - ☐ 4th
 - ☐ 14th
 - ☒ No output wire will be activated Page 163
148. Half-Adder Logic circuit contains 2 XOR Gates
- ☐ True
 - ☒ False Page 135
149. The decimal “8” is represented as using Gray-Code.
- ☐ 0011
 - ☒ 1100 page 36
 - ☐ 1000
 - ☐ 1010
150. NOR Gate can be used to perform the operation of AND, OR and NOT Gate
- ☐ FALSE
 - ☒ TRUE Page 50
151. In ANSI/IEEE Standard 754 “Mantissa” is represented by 32-bits bits
- ☐ 8-bits
 - ☐ 16-bits
 - ☒ 32-bits Page 24
 - ☐ 64-bits
129. Caveman number system is Base _5 number system
- ☐ 2
 - ☒ 5 Page 11
 - ☐ 10
 - ☐ 16
152. The Extended ASCII Code (American Standard Code for Information Interchange) is a.....code
- ☒ 2-bit

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- ✚ 7-bit
- ✚ 8-bit Page 38
- ✚ 16-bit
- 153. NOR gate is formed by connecting.....
 - ✚ OR Gate and then NOT Gate Page 47
 - ✚ NOT Gate and then OR Gate
 - ✚ AND Gate and then OR Gate
 - ✚ OR Gate and then AND Gate
- 154. Generally, the Power dissipation of.....devices remain constant throughout their operation.
 - ✚ TTL Page 65
 - ✚ CMOS 3.5 series
 - ✚ CMOS 5 Series
 - ✚ Power dissipation of all circuits increases with time.
- 155. When the control line in tri-state buffer is high the buffer operates like a..... gate
 - ✚ AND
 - ✚ OR
 - ✚ NOT Page 196
 - ✚ XOR
- 156. All the ABEL equations must end with.....
 - ✚ “. “(a dot)
 - ✚ “\$ “(a dollar symbol)
 - ✚ “; “(a semicolon) Page 201
 - ✚ “endl “(keyword “endl”)
- 157. Circuits having a bubble at their outputs are considered to have an active-low output.
 - ✚ True Page 128
 - ✚ False
- 158. The ABEL symbol for “XOR” operation is
 - ✚ \$ page 210
 - ✚ #
 - ✚ !
 - ✚ &
- 159. The main use of the Multiplexer is to
 - ✚ Select data from multiple sources and to route it to a single Destination Page 167
 - ✚ Select data from Single source and to route it to a multiple Destinations
 - ✚ Select data from Single source and to route to single destination
 - ✚ Select data from multiple sources and to route to multiple destinations
- 160. A logic circuit with an output $X = \bar{A} B C + A \bar{B}$ consists of.....
 - ✚ two AND gates, two OR gates, two inverters
 - ✚ three AND gates, two OR gates, one inverter
 - ✚ two AND gates, one OR gate, two inverters
 - ✚ two AND gates, one OR gate
- 161. The Unsigned Binary representation can only represent positive binary numbers
 - ✚ True Page 21

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- ✚ False
162. A BCD to 7-Segment decoder has
- ✚ 3 inputs and 7 outputs
 - ✚ 4 inputs and 7 outputs **Page 103**
 - ✚ 7 inputs and 3 outputs
 - ✚ 7 inputs and 4 outputs
163. The 4-variable Karnaugh Map (K-Map) has.....cells for min or max terms
- ✚ 4
 - ✚ 8
 - ✚ 12
 - ✚ 16 **Page 90**
164. The PROM consists of a fixed non-programmable.....Gate array configured as a decoder.
- ✚ AND **Page 182**
 - ✚ OR
 - ✚ NOT
 - ✚ XOR
165. In ABEL the variable „A“ is treated separately from variable „a“
- ✚ True **Page 201**
 - ✚ False
 - ✚ The ABEL notation equivalent to Boolean expression $A+B$ is: $A \& B$
 - ✚ $A! B$
 - ✚ $A \# B$ **Page 201**
 - ✚ $A \$ B$
166. Which one is true:
- ✚ Power consumption of TTL is higher than of CMOS **Page 61**
 - ✚ Power consumption of CMOS is higher than of TTL
 - ✚ Both TTL and CMOS have same power consumption
 - ✚ Power consumption of both CMOS and TTL depends on no. of gates in the circuit.
167. The decimal equivalent of the binary number “10011” is
- ✚ 19 **(According to rule)**
 - ✚ 99
 - ✚ 29
 - ✚ None of given options
168. “74ALS” stands for
- ✚ Advanced Low-frequency Schottky TTL
 - ✚ Advanced Low-dissipation Schottky TTL
 - ✚ Advanced Low-Power Schottky TTL **Page 61**
 - ✚ Advanced Low-propagation Schottky TTL
169. An adder circuit can be used to perform subtraction operation
- ✚ True **Page 146**
 - ✚ False
170. The four outputs of two 4-input multiplexers, connected to form a 16-input multiplexer, are connected together through a 4-input gate
- ✚ AND

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-  OR Page 171-172
-  NAND
-  XOR
171. The Programmable Array Logic (PAL) has AND array and a OR array
-  Fixed, programmable
-  Programmable, fixed Page 182
-  Fixed, fixed
-  Programmable, programmable
172. Sequential circuits have storage elements
-  True Page 218
-  False
173. GAL can be reprogrammed because instead of fuses.....logic is used in it
-  E²CMOS Page 191
-  TTL
-  CMOS+
-  None of the given options
174. Which of the following is a volatile memory
-  PROM
-  DRAM
-  EPROM
-  EEPROM
175.is used when the output is connected back to the input of PAL or if the output pin is used as an input only.
-  Combinational Input/Output
-  Combination Output
-  Combinational Input
-  Programmable polarity
176. The AND Gate performs a logical.....function.
-  Addition
-  Subtraction
-  Multiplication
-  Division
177. The Adjacent 1s detector accepts 4-bit inputs. Ifadjacent 1s are detected in the input, the output is set to high.
-  2
-  4
-  1
-  0
178. In the keyboard encoder, how many times per second does the ring counter scan the keyboard?
-  600 scans/second
-  625 scans/second
-  650 scans/second
-  700 scans/second
179. The FAST Model Page Access allows.....memory read and access times when reading successive data values stored in consecutive locations on the same row

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- ☐ Slow
 - ☒ Faster
 - ☐ Medium
 - ☐ Modern
180. GAL can be reprogrammed as instead of fuses E²CMOS logic is used which can be programmed to connect a.....with a.....
- ☐ Column, row
 - ☒ Row, column
 - ☐ Column, column
 - ☐ Row, row
181. Which of the following output equations determines the output of the state machine?
- ☐ MIN = Q0Q1
 - ☒ MAX = Q0Q1EN
 - ☐ MIN = Q0Q1EN
 - ☐ MAX = Q1EN
182. The maximum value, represented by a single hexadecimal digit is.....
- ☐ "E"
 - ☒ "F"
 - ☐ "G"
 - ☐ "H"
183. If the voltage drop across the active load is 0 volts due to absence of current the comparator output is a.....
- ☐ 0
 - ☒ 1
184. The Static Ram (SRAM) is non-volatile and is not a.....density memory as a latch is required to store a single bit of information.
- ☐ Low
 - ☒ High
 - ☐ Medium
 - ☐ Hot
185. DE Morgan's two theorems prove the equivalency of the NAND and.....gates and the NOR and.....gates respectively.
- ☒ Negative-OR, Negative-AND
 - ☐ Negative-AND, Positive-OR
 - ☐ Positive-OR, Negative-AND
 - ☐ Positive-OR, Positive-AND
186. Two signals.....and.....provide the timing inputs to the state machine.
- ☐ NSSR and EWSR
 - ☐ LTIME and STIME
 - ☒ PTIME and QTIME
 - ☐ NSGrn and NSYel
187. The 74HC163 is a 4-bit synchronous counter, it has.....data outputs pins.
- ☐ 2
 - ☒ 4
 - ☐ 6
 - ☐ 8

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188. PLDs have In-system Programming (ISP) capability that allows the.....to be programmed after they have been installed on a circuit board.

- ☐ PLAs
- ☐ PALs
- ☒ PLDs
- ☐ EPROM

189. The CONSTATE.CLK = Clock is used to indicate that the.....state variables change on a clock transition.

- ☒ CONSTATE
- ☐ FLOOR
- ☐ MOTION
- ☐ OPEN

190. Two types of memories namely the first in-first out (FIFO) memory and last in first out (LIFO) are implemented using.....

- ☒ Shift Register
- ☐ Circular Buffers
- ☐ Ring Buffers
- ☐ Reduce Register

191. The normal data inputs to a flip-flop (D, S and R, J and K, T) are referred to as.....inputs.

- ☐ Sequential
- ☐ Asynchronous
- ☒ Synchronous
- ☐ Combinational

192. For a down counter that counts from (111 to 000), if current state is “101” the next state will be.....

- ☐ 111
- ☐ 110
- ☐ 010
- ☒ None of the given

193. The.....gate and.....gate implementation connected at the B input of the 4-bit Adder is used to allow Complemented or Un-complemented B input to be connected to the Adder input.

- ☐ AND, NOR
- ☐ AND, NOT
- ☐ AND, OR
- ☒ XOR, NAND

194. The Synchronous c allows the Synchronous SRAM to reader or writer up to.....location(s) using a single address.

- ☐ One
- ☐ Two
- ☐ Three
- ☒ Four

195. In NAND based S-R latch, output of each.....gate is connected to the input of the other.....gate.

- ☐ NOR, NAND
- ☒ NAND, NOR

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✚ NOR, NOR

✚ NAND, NAND

196. Implementing the Adjacent 1s detector circuit directly from the function table based on the SOP form requires.....gates for the 8 product terms (minterms) with an 8-input OR gate.

✚ 8 OR

✚ 8 AND

✚ 8 XOR

✚ 8 NOR

197. 8-bit parallel data can be converted into serial data by using.....multiplexer.

✚ 4-to2

✚ 8-to-1

✚ 4-to-4

✚ 8-to-4

198. The.....input overrides the.....input.

✚ Asynchronous, synchronous

✚ Synchronous, asynchronous

✚ Preset input (PRE), Clear input (CLR)

✚ Clear input (CLR), Preset input (PRE)

199. A SOP expression can be implemented by an.....combination of gates.

✚ OR-XOR

✚ AND-NAND

✚ AND-OR

✚ XOR-NOR

200. The 64-cell array organized as 8×8 cell array is considered

✚ As an 64byte memory

✚ As a 16byte memory

✚ As an 8byte memory

✚ As a 4byte memory

201. The terminal count of a 4-bit binary counter in the UP mode is.....

✚ 1100

✚ 0011

✚ 1111

✚ 0000

202. An asynchronous Down-counter is implemented (Using J-K flip-flop) by connecting.

✚ Q output of all flip-flops to clock input of next flip-flops

✚ Q' output of all flip-flops to clock input of next flip-flops

✚ Q output of all flip-flops to J input of next flip-flops

✚ Q' output of all flip-flops to K input of next flip-flops

203. Memory is arranged in.....

✚ Linear fashion

✚ Two-dimensional manner

✚ Three-dimensional manner

✚ Random fashion

204. Subtractors also have output to check if 1 has been.....

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- ☒ Primed
 - ☒ Shifted
 - ☒ Complemented
 - ☒ Borrowed
205. The test Vector definition defines the test vectors for all the three counter inputs and.....counter outputs.
- ☒ One
 - ☒ Two
 - ☒ Three
 - ☒ Four
206. A multiplexer with a register circuit converts
- ☒ Serial data to parallel
 - ☒ Parallel data to serial
 - ☒ Serial data to serial
 - ☒ Parallel data to parallel
207. A decade counter can be implemented by truncating the counting sequence of a MOD-20 counter.
- ☒ True
 - ☒ False
208. The n flip-flop store.....states.
- ☒ 1
 - ☒ 2^n
 - ☒ 0
 - ☒ $2^{(n+1)}$
209. The S-R latch has two inputs, therefore.....different combinations of inputs can be applied to control the operation of the S-R latch.
- ☒ Two
 - ☒ Four
 - ☒ Eight
 - ☒ Sixteen
210. Why demultiplexer is called a data distributor?
- ☒ The input will be distributed to one of the outputs
 - ☒ The input will be selected for the output
 - ☒ The output will be distributed to one of the inputs
 - ☒ Single input to single output
211. When the transmission line is idle in an asynchronous transmission.
- ☒ It is set to logic low
 - ☒ It is set to logic high
 - ☒ It remains in previous state
 - ☒ State of transmission line is not used to start transmission
212. UVERPROM is stands for
- ☒ Ultra-Variant
 - ☒ Ultra-Vibrant
 - ☒ Ultra-Voilet
 - ☒ Ultra-Visible
213. In memory write cycle, the time for which the WE single remains active is known as the.....

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- ☐ Write address setup
 - ☒ Write pulse width
 - ☐ Write delay width
 - ☐ Write date time
214. The outputs of SR latches in elevator state machine are feed back to the.....gate array for connection to the D-flipflops.
- ☐ NOT
 - ☒ AND
 - ☐ OR
 - ☐ XOR
215. PALs tend to execute.....logic.
- ☐ SPD
 - ☒ SOP
 - ☐ SAC
 - ☐ SAP
216. The ROM used by a computer is relatively.....as it stores few byers of code used to Boot the computer system on power up.
- ☒ Small
 - ☐ Large
 - ☐ Heavy
 - ☐ High
217. Which single must remain valid in memory write cycle after data is applied at the data input lines and must remain valid for a minimum time duration t_{WD} ?
- ☐ -CS
 - ☒ -WE
 - ☐ W
 - ☐ OE
218. You have to choose suitable option when your timer will reset by considering this given code:
- ```
TRSTATE.CLK = clk;
TMRST: = (TRSTATE == NSY2) # (TRSTATE == EWY2);
```
- ☒ NSY2 or EWY2
  - ☐ NSSR or TMRST
  - ☐ EWSR or NSRED
  - ☐ EWRed or EWYel
219. A NOR based S-R latch is implemented using.....gates instead of.....gates.
- ☐ XOR, NAND
  - ☐ NOR, XOR
  - ☒ NOR, NAND
  - ☐ OR, XOR
220. Implementation of latch is required almost.....transistor.
- ☐ Two
  - ☐ Four
  - ☒ Six
  - ☐ Eight

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221. In distributed mode, for a  $1024 \times 1024$  DRAM memory and a refresh cycle of 8 msec, each of the 1024 rows has to be refreshed in.....when distributed refresh is used.
- ☐ 4.8 microsec
  - ☐ 5.9 microsec
  - ☒ 7.8 microsec
  - ☐ 5.5 microsec
222. The NOR logic gate is the same as the operation of the.....gate with an inverter connected to the output.
- ☐ AND
  - ☒ NAND
  - ☐ OR
  - ☐ NOT
223. For a standard SOP expression, a.....is placed in the cell corresponding to the product term (Minterm) present in the expression.
- ☐ 0
  - ☒ 1
  - ☐ x (don't care condition)
  - ☐ Any of given option depending on SOP term
224. Select the mode of programming in which GAL 16V8 can be programmed:
- ☐ Simple Mode
  - ☐ Complex Mode
  - ☐ Registered Mode
  - ☒ All of the given
225. Divided-by-32 counter can be achieved by using,
- ☐ Flip-Flop and DIV 10
  - ☐ Flip-Flop and DIV 16
  - ☒ Flip-Flop and DIV 32
  - ☐ DIV 16 and DIV 32
226. The next state table for REQ1, FLOOR1 and OPEN inputs indicates that the.....can be pressed at any time either on the first floor or the second floor in elevator.
- ☐ REQ0
  - ☐ OPEN
  - ☒ REQ1
  - ☐ FLOOR1
227. Consider  $A=1$ ,  $B=0$  and  $C=1$ . A,B and C represent the input of three bit NAND gate, the output of the NAND gate will be.....
- ☐ Zero
  - ☒ One
  - ☐ Undefined
  - ☐ No output as input is invalid
228. A 4-bit binary up/down counter is in the binary state of zero. The next state in the down mode is:
- ☐ 0001
  - ☐ 1000
  - ☐ 1110

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229. Adding two octal numbers “36” and “71” result in.....  
☐ 213  
☐ 123  
☒ 127  
☐ 345
230. The ABEL input file can use a.....instead of the equation to specify the Boolean expression.  
☒ Truth Table  
☐ State Diagram  
☐ Karnaugh Map  
☐ Logic Circuit
231. The domain of the expression  $AB'CD + AB' + C'D + B$  is  
☐ A and D  
☒ B only  
☐ A, B and D  
☐ None of the given
232. If the number of samples that are collected is reduced by half, the reconstructed signal will be.....from/to the original.  
☐ Different  
☒ Same  
☐ Equal  
☐ Opposite
233. In DRAM read cycle R/W signal is activated to read data which is made available on the.....data line.  
☐ D(IN)  
☒ D(OUT)  
☐ D(AB)  
☐ D(INT)
234. In case of cascading integrated circuit counters. the enable inputs and RCO of the integrated circuit counters allow cascading of multiple counters together.  
☒ True  
☐ False
235. Implementation of the FIFO buffer in.....is usually takes the from of a circular buffer.  
☒ RAM  
☐ ROM  
☐ PPRM  
☐ Flash memory
236. As data values are written or read from the RAM stack pointer register increments or decrements its contents always pointing to the stack.....  
☐ Bottom  
☒ Top  
☐ Down  
☐ Vertex
237. Which one flip-flop has an invalid output state?  
☒ T

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- ☐ JK  
☒ SR  
☐ D
238. The output of a NAND gate is.....when all the inputs are one.  
☒ Zero  
☐ One  
☐ Available  
☐ Unavailable
239. The transition table is very similar to the.....table.  
☐ Truth  
☒ State  
☐ Transition  
☐ None of the given
240. Consider the sum of weight method for converting decimal into binary value.....is the highest weight for 411.  
☐ 64  
☐ 128  
☒ 256  
☐ 512
241. Canonical form is a unique way of representing.....  
☒ SOP  
☐ Minterm  
☐ Boolean Expression  
☐ POS
242. ....counters as the name indicates are not triggered simultaneously.  
☐ Asynchronous  
☒ Synchronous  
☐ Positive-Edge triggered  
☐ Negative-Edge triggered
243. Cin is part of.....adder.  
☐ Half  
☒ Full  
☐ Single  
☐ Double
244. Flash memory operation are classified into.....different operation.  
☒ Two  
☐ Three  
☐ Four  
☐ Five
245. In 8-input multiplexer, the two outputs are connected through a/an\_\_\_gate.  
☐ AND  
☒ OR  
☐ NOT  
☐ NOR
246. ....Device dissipate varying amount of power depending upon the frequency of operation.

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- ☐ TTL
- ☒ CMOS
- ☐ Storage
- ☐ Peripheral

247. Boolean Addition operation is performed by a(an).....gate.

- ☐ AND
- ☒ OR
- ☐ XOR
- ☐ NAND

248. The maximum decimal number that can be represented using the 64-bit unsigned representation is.....

- ☐  $2^{63}$
- ☒  $(2^{64})-1$
- ☐  $(2^{64}+$
- ☐  $2^{64}$

249. In 16-bit ALU, The G output is activated if the 4-bit unit generate a Carry.....irrespective of Carry.....

- ☐ In, in
- ☐ In, out
- ☒ Out, in
- ☐ Out, out

250. In Cascading Priority Encoders, the EO output is connected to the EI input of the encoder which handles\_\_\_\_\_.

- ☐ Lower priority outputs
- ☐ Higher priority outputs
- ☒ Lower priority inputs
- ☐ Higher priority inputs

251. Which of the following is the example of comparator?

- ☐ OR
- ☐ AND
- ☐ XOR
- ☒ XNOR

252. The Adjacent 1 S Detector accepts 4-bits input. If \_\_\_\_\_ adjacent 1S are detected in the input, the output is set to high.

- ☐ 2
- ☒ 4
- ☐ 1
- ☐ 0

253. De Morgan's two theorems prove the equivalency of the NAND and \_\_\_\_\_ gates and the NOR and \_\_\_\_\_ gates respectively.

- ☐ Negative-OR, Negative-AND
- ☒ Negative-AND, Negative-OR
- ☐ positive-OR, Negative-AND



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- positive-AND, Negative-OR
254. Adding two octal numbers “36 and 71” result in \_\_\_\_.
- 213
  - 123
  - 127**
  - 345
255. Any of the \_\_\_\_ forms of the Karnaugh Map can be used to simplify Boolean expressions
- Three
  - Four
  - Two**
  - Five
256. Quine-McCuskey and K-Map methods are used for \_\_\_\_ of Boolean expression.
- Multiplication
  - Addition
  - Simplification**
  - Subtraction
257. The number “1259” may belong to \_\_\_\_ number system.
- Binary number system
  - Octal or Decimal system
  - Decimal or Hexadecimal system**
  - Binary or Hexadecimal system
258. The series of TTL chips are characterized by their \_\_\_\_.
- Switching Speed only**
  - Power Dissipation only
  - Both
  - None of these
259. All ABEL statements must end with \_\_\_\_.
- :
  - ,
  - ;
  - “
260. In sequential circuit memory elements are connected with \_\_\_\_.
- Logical circuit
  - Clock**
  - Feedback path
  - External Event
261. In the 32-bit Single Precision Floating Point format, the exponent value \_\_\_\_ is reserved to represent infinity exponents.
- 98
  - 99
  - 255**
  - 256

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262. The \_\_\_\_ output has the output of the OR gate connected through an XOR gate to the tri-state buffer.

- + Combinational
- + Combinational input/
- + PLA
- + **None** Programmed Polarity

263. The limitation in implementation of parallel binary address is known as \_\_\_\_.

- + Delay
- + Carry propagation
- + **Carry input**
- + Carry output

264. The Gray code is different from the unsigned binary code because \_\_\_\_.

- + **Successive value of Gray code by only one bit**
- + Gray code is positional code
- + Gray code not support negative values
- + Gray code ranges from "0" to "9"

265. Removing the NOT gate at the output of the NOR gate results in an \_\_\_\_.

- + **OR gate** AL-JUNAID INSTITUTE OF GROUP
- + NAND gate
- + AND gate
- + NOT gate

266. Portable devices that run on batteries use \_\_\_\_ circuit that have low power dissipation.

- + Series
- + **Integrated**
- + Parallel
- + Electric

267. \_\_\_\_ is a single input gate

- + AND
- + **OR**
- a. NOT
- b. XOR

268. To represent in digital value, the number of digits (0s and 1s) that represent a quantity is \_\_\_\_ to the range of values that are to be represented.

- + **Equal**
- + Greater

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- ✚ Lesser  
✚ Not equal
269. BCD code of 16 is \_\_\_\_.
- a. 10001
  - b. **00010001**
  - c. 00010111
  - d. 01110001
270. In Odd parity generator circuit which gate is used to detect parity errors?
- a. OR
  - b. NOR
  - c. **XOR**
  - d. AND
271. The measurable values generally change over a
- ✚ Specified period
  - ✚ Discrete range
  - ✚ Time
  - ✚ **Continuous range**
272. \_\_\_\_ uses E2CMOS technology which is Electrically Erasable CMOS instead of Bipolar technology and fusible links.
- ✚ PAL
  - ✚ **GAL**
  - ✚ PLA
  - ✚ PROM
273. When the number 29 is represent on 7-segment display, which BCD input is represented on LSD display unit?
- ✚ 1000
  - ✚ **1001**
  - ✚ 1010
  - ✚ 1100
274. How many of enable inputs is(are) active-low in 74xx138 3 to 8 Decoder?
- a. One
  - b. Two
  - c. **Three**
  - d. Four
275. The simplified expression using either of the two K-maps are \_\_\_\_.
- ✚ Similar
  - ✚ Different

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☒ **Identical**

☐ None-identical

276. Which of the following expression in the product of sums form?

☐  $(A + B)(C + D)$

☐  $(AB)(CD)$

☐  $AB(CD)$

☒  **$AB + CD$**

277. CMOS technology is characterized by low power dissipation with \_\_\_\_\_ switching speeds.

☒ **a. Slow**

☐ b. Fast

☐ c. Average

☐ d. Medium

278. GAL Two 2-bit comparator circuits can be connected to form single 4-bit comparator

☒ **True**

☐ False

279. High level Noise Margins ( $V_{NH}$ ) of CMOS 5 volt series circuits is

☐ a. 0.2 V

☐ b. 0.5 V

☒ **c. 0.9 V**

☐ d. 3.3 V

280. The output of the expression  $F = A + B + C$  will be Logic \_\_\_\_\_ when  $A = 0$ ,  $B = 1$ ,  $C = 1$ . the symbol "+" here represents OR Gate

☐ Undefined

☒ **One**

☐ Zero

☐ 10(binary)

281. If an active-HIGH S-R latch has a 0 on the S input and a 1 on the R input and then the R input goes to 0, the latch will be \_\_\_\_\_.

☒ **a. SET**

☐ b. RESET

☐ c. Clear

☐ d. Invalid

282. 3.3 v CMOS series is characterized by \_\_\_\_\_ and \_\_\_\_\_ as compared to the 5 v CMOS series.

☐ a. Low switching speeds, high power dissipation

☐ b. Fast switching speeds, high power dissipation

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- c. Fast switching speeds, very low power dissipation (page61)**
- d. Low switching speeds, very low power dissipation
283. The binary value "1010110" is equivalent to decimal \_\_\_\_\_
- a. 86**
- b. 87
- c. 88
- d. 89
284. The \_\_\_\_\_ Encoder is used as a keypad encoder.
- a. 2-to-8 encoder
- b. 4-to-16 encoder
- c. BCD-to-Decimal
- d. Decimal-to-BCD Priority**
285. How many data select lines are required for selecting eight inputs?
- a. 1
- b. 2
- c. 3**
- d. 4
286. What is the output expression of segment 'b' implementation in BCD to 7-segment decoder?
- a.  $B + C'D' + CD$
- b.  $B' + C'D' + CD$**
- c.  $B + C'D' + C'D$
- d.  $B' + C'D' + CD'$
287. 2-input, 8-bit Multiplexer, by setting the S input to logic \_\_\_\_\_ the \_\_\_\_\_ inputs of both the multiplexers are selected.
- a. Low
- b. High, B**
- c. Low, C
- d. Low, C
288. When two or more products terms are assumed by Boolean addition, the result is a \_\_\_\_\_
-  POS
-  **SOP**
-  Boolean
-  Simplified
289. Tri-State Buffer is a \_\_\_\_\_ gate with a control line that disconnects the
- a. OR
- b. AND
- c. NAND

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d. NOT

290. The 4-bits 2's complement representation of "7" is \_\_\_\_\_
- ☐ 0111
  - ☐ 1111
  - ☒ 1001
  - ☐ 0110
291. \_\_\_\_\_ and \_\_\_\_\_ are the steps of the Quine-McCluskey.
- ☐ Draw a table and make groups
  - ☐ Write binary codes and write SOP
  - ☒ Find prime implicants and select minimal set of the prime implicants.
  - ☐ None of the given
292. The binary number 1011,101 has an Integer part represented by \_\_\_\_\_ and a fraction part \_\_\_\_ separated by a decimal point.
- ☒ a. 1011,101
  - ☐ b. 101,1011
  - ☐ c. 101,1101
  - ☐ d. 10111,11
293. Subtractors also have output to check if 1 has been \_\_\_\_\_
- ☐ a. Primed
  - ☐ b. Shifted
  - ☐ c. Complemented
  - ☒ d. Borrowed
294. The \_\_\_\_\_ description is used to simulate the logic circuit and verify its operation.
- ☐ Test file
  - ☐ Declaration
  - ☐ Logic
  - ☒ Test vector
295. How many outputs can an integrated circuit comparator have?
- ☐ One
  - ☐ Two
  - ☒ Three
  - ☐ Four
296. Which of the following is not the correct method of grouping?
- ☐ Along adjacent columns
  - ☐ On comers
  - ☒ Diagonally
  - ☐ Along adjacent rows
297. The output of the expression  $F=A.B.C$  will be logic \_\_\_\_\_

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when  $A=1, B=0, C=1$ .

- a. Undefined
- b. One
- c. Zero
- d. No output as input is valid

298. The \_\_\_\_ gate and \_\_\_\_ gate implementation connected at the B, input of the 4-bit Adder is used to allow complemented or Un-Complemented B input to be connected to the Adder input.

- ✚ AND,NOR
- ✚ AND,NOT
- ✚ AND,OR
- ✚ XOR,NAND

299. In the 32-bit Single Precision Floating point format, the exponent value \_\_\_\_ is reserved infinity exponent.

- i. 98
- ii. 99
- iii. 255
- iv. 256

300. The Boolean expression  $(AB'CS')$  is used

- ✚ A sum term
- ✚ A product term
- ✚ A literal term
- ✚ A max term

301. The product of an XOR gate is zero(0), when \_\_\_\_\_

- I. All the inputs are zero
- II. Any of the inputs is zero
- III. An of the inputs is one
- IV. All the inputs are one

- ✚ I only
- ✚ IV only
- ✚ I and IV only
- ✚ II and III

302. \_\_\_\_\_ methods are used to Convert Decimal fractions to Binary.

- i. 1
- ii. 2
- iii. 3
- iv. 4

303. To display the number\_\_\_\_ the BCD number 0010 representing the MSD is applied at the inputs of the BCD to 7-segment display circuit connected to the MSD &-Segment Display digit

- ✚ 19

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✚ 29

✚ 39

✚ 49

304. If two numbers in BCD representation generate an invalid BCD number then the binary \_\_\_ is added to the result
- a. 1001
  - b. 0110
  - c. 1111
  - d. 1100
305. Both the multiplexers are selected simultaneously when \_\_\_\_\_ is set to logic \_\_\_\_\_ in 2-inputs, 8-bit Multiplexer.
- a. G, Low
  - b. G, High
  - c. S, Slow
  - d. S, High
306. Function labels required to represent the input/output combinations for each segment in 7-segment display
- a. 5
  - b. 7
  - c. 8
  - d. 10
307. Multiplexers are also known as \_\_\_\_\_
- a. Date distributors
  - b. Data selectors
  - c. Data manipulators
  - d. Data setters
308. The PLA can be programmed to give an output of constant \_\_\_\_\_ or \_\_\_\_\_
- a. 0,1
  - b. 1,2
  - c. 2,3
  - d. 0,0
309. The look-ahead carry circuits \_\_\_\_\_
- a. Add 1 to complemented inputs
  - b. Reduce propagation delay
  - c. Increase ripple delay
  - d. Determine sign and magnitude
310. Which of the following gates has the outputs 1 if and only if at least one input is 1?
- ✚ AND



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- ☐ NOR
- ☐ NAND
- ☒ OR

311. A SOP expression can be implemented by on \_\_\_\_ combination of gates.
- ☐ OR-XOR
  - ☐ AND-NAN
  - ☒ AND-OR
  - ☐ XOR-NOR
312. The carry, instead of rippling through the 4-bits of the individual ALU circuit, has to propagate through \_\_\_\_ ALU units in 16-bit ALU.
- ☐ Two
  - ☒ Four
  - ☐ Nine
313. Digital circuits operate with \_\_\_\_\_ voltage value(s)
- a. 1
  - ☒ b. 2
  - c. 3
  - d. 4
314. Which of the following is the octal equivalent of 28 decimal numbers?
- ☐ 31
  - ☐ 32
  - ☐ 33
  - ☒ 34
315. To determine the seven expressions for each of the seven outputs in 7-segment display, seven \_\_\_\_ variable Karnaugh maps are used.
- ☐ 3
  - ☒ 4
  - ☐ 5
  - ☐ 2
316. The output of a NAND Gate is \_\_\_\_ when all the inputs are one.
- ☐ Zero
  - ☐ One
  - ☐ Available
  - ☐ Not available
317. The \_\_\_\_\_ is the slowest and consumes more power.
- ☒ a. Standard TTL
  - b. Schottky TTL
  - c. Advanced Schokottky TTL
  - d. Low-Power Schottky TTL
318. The between expression  $X-AB+CD$  represents

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- ✚ Two ORs ANDed together
  - ✚ A 4-input AND gate
  - ✚ Two ANDs ORed together
  - ✚ An exclusive-OR
319. The expression  $F = A + B + C$  describes the operation of three bits \_\_\_\_ Gate.
- ✚ OR
  - ✚ AND
  - ✚ NOT
  - ✚ NAND
320. Which one of the following is NOT a valid rule of Boolean Algebra?
- ✚  $A + 1 = 1$
  - ✚  $A = A'$
  - ✚  $AA = A$
  - ✚  $A + 0 = A$
321. A 5-Variable Karnaugh map has
- ✚ Sixteen cells
  - ✚ Thirty-two cells
  - ✚ Sixty-four cells
  - ✚ None of these
322. \_\_\_\_ is invalid number of cells in a single group formed by the adjacent cells K-map
- ✚ 2
  - ✚ 8
  - ✚ 12
  - ✚ 16
323. In 32-bit Single –Precision floating point format representation the range of exponent value is from \_\_\_\_ to \_\_\_\_
- ✚ +127 to -126
  - ✚ +127 to -254
  - ✚ +128 to -254
  - ✚ +256 to -255
324. \_\_\_\_ has the fastest switching speed and low power requirements
- ✚ Advanced low power Schottky
  - ✚ Fast TTL
  - ✚ Standard TTL
  - ✚ Schottky TTL
325. A standard sum term is equal to zero for only one combination of \_\_\_\_ values.
- ✚ Literal
  - ✚ Maximum
  - ✚ Variable

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- None
326. The output of an AND gate with three inputs, A, B, and C, is HIGH when \_\_\_\_\_.
- A=1, B=1, C=0
  - A=1, B=1, C=1
  - A=1, B=0, C=1
  - A=0, B=1, C=1
327. The output of an exclusive-OR gate is HIGH if \_\_\_\_\_.
- All Input or Unequal
  - All inputs are DON'T care
  - All inputs are HIGH
  - All inputs are low
328. On a Karnaugh map, grouping the 0s produces \_\_\_\_\_.
- A POS expression
  - A SOP expression
  - AND-OR Logic
  - NAND-NOR Logic
329. The simplified expression is presented in a \_\_\_\_\_ format.
- Function Table
  - Logic Table
  - Truth Table
  - Data Table
330. The boolean expression  $X = AB + CD$  represents.
- two ORs ANDed together
  - a 4-input and gate
  - two ands ored together
  - an exclusive- or
331. Which of the following is the example of comparator?
- SPD
  - SOP
  - SAC
  - SAP